

# Arhaan Penwala

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## EDUCATION

### Vellore Institute of Technology

B.Tech Computer Science and Engineering (AI and Robotics), CGPA: **9.13/10.0**

Chennai, India

Expected May 2027

## EXPERIENCE

### Co-Founder and CTO

Nov '24 – Present

V Help | Flutter, Supabase (PostgreSQL, Deno Edge Functions), Firebase, Azure

VIT Chennai

- Built and operate the backend of a Flutter campus super-app serving **1,000+** students, as one of 2 core engineers
- Merged two Firebase projects under one Supabase identity with a custom JWT exchange, at **zero re-registration**
- Closed a privilege-escalation surface by moving **24** write paths off the client into server-authorized Edge Functions

### Software Development Engineer Intern

Dec '25 – Feb '26

BPO Integra India Private Limited

Remote

- Cut response times on **3** production API endpoints by collapsing N+1 queries and adding composite MySQL indexes
- Shipped a reusable React component library and Node.js/Express REST services for a live production site

### AI Research Intern

May – Jun '25

The Indian Hotels Company Limited (Taj Hotels, Tata Group)

Mumbai, India

- Built a Python cost model over **5** enterprise LLM vendors and **15** hospitality use cases, projecting a **35%** spend reduction that drove leadership's vendor selection
- Benchmarked the 5 vendor chatbots on latency and response quality to derive the model's per-use-case parameters

## PROJECTS

MutaFix | LangGraph, pydantic-ai, Gemini API, tree-sitter, Docker, pytest

Aug '26

- Measured a LangGraph repair agent at **93.0% strict\_pass@3** ( $\pm 2.5$  SE) over **105 challenges**  $\times$  **3 seeds**, by generating **1,084** contamination-free bugs through deterministic tree-sitter AST mutation of well-tested Python repositories
- Exposed **5 of 298** passing runs that left the injected bug fully intact, by hand-verifying every non-exact repair against executable probes; **pass@k** alone scores all five as successes
- Executed every LLM-generated patch in a no-network Docker sandbox; the full benchmark reproduces from a seed in one command

Gait-Based Deepfake Detection | PyTorch, MediaPipe, Grad-CAM, Scikit-learn, Jetson Nano

Jun '25 – Present

- Detected face-swap deepfakes from gait alone with a CNN + BiLSTM + Transformer hybrid, **no facial features**
- Reached **94.95% AUC-ROC** and 87.0% accuracy under leave-one-subject-out cross-validation, ported to **Jetson Nano**

PolarisGCS | FastAPI, TimescaleDB, Redis, React/TypeScript, PySide6, Kotlin, Docker

Feb '26 – Present

- Streamed telemetry from **15 concurrent MAVLink vehicles** (ArduPilot SITL) via Redis pub/sub fan-out and TimescaleDB retention
- Defined one wire protocol across React web, PySide6 desktop, and Kotlin Android clients: **zero backend changes**

## TECHNICAL SKILLS

**Languages:** Python, Java, C/C++, JavaScript/TypeScript, SQL, Kotlin, Dart

**AI & LLM Engineering:** LangGraph, pydantic-ai, Gemini API, agentic state machines, structured output schemas, prompt design, agent evaluation and benchmark design (pass@k, SEM), tree-sitter AST analysis, LLM cost modeling

**Machine Learning:** PyTorch, Scikit-learn, XGBoost, Pandas, NumPy, MediaPipe, Grad-CAM, CNN/BiLSTM/Transformer architectures

**Backend & Infrastructure:** FastAPI, Node.js, Express.js, REST APIs, WebSockets, Redis Pub/Sub, Docker, PostgreSQL/TimescaleDB, MySQL, MongoDB, Supabase, Firebase, Deno Edge Functions, AWS, Linux, GitHub Actions, React.js, Flutter

## ACHIEVEMENTS AND PUBLICATIONS

**2nd Place** out of **500+** teams, Hack N Droid '25 | **Publication:** gait dataset, IEEE DataPort, DOI 10.21227/ng5-b637